



Felipe Coutinho

AI Engineer

AI Engineer with a Master of Science in Engineering Physics and 4 years of combined research and industry experience. Owns AI systems end to end, from framing the problem and building the data pipeline through model training, evaluation, and deployment. Track record spanning LLM and agentic automation, multimodal AI, and computer vision, delivered for research partners and early-stage startups.

📍 Porto, Portugal ✉ [fcoelho.mrc \(at\) gmail \(dot\) com](mailto:fcoelho.mrc@gmail.com)
🌐 [fcoelho.mrc](#) (LinkedIn) 🐙 [fcoelho.mrc](#) (GitHub)

Skills

Python Rust PyTorch TensorFlow Scikit-learn Hugging Face
LoRA OpenCV torchvision albumentations vLLM SmolVLM
Qwen LangChain LangGraph OpenAI API AWS Google Cloud
Kubernetes Terraform PostgreSQL MongoDB Pinecone Docker
Slurm Linux Graphana FastAPI Flask MLFlow
GitHub Actions Git

Work INESC TEC

AI Researcher

10/2024 – Present

- Owning end to end a multimodal AI project with the University of Cambridge: designed the architecture, curated and integrated TB-scale datasets, and trained models predicting chemotherapy response from pre-treatment CT scans, reaching 70% accuracy on indicators normally available only post-surgery; collaborating with clinical stakeholders in hospitals to frame the question and validate results.
- Ranked top-5 in the MICCAI 2025 CHIMERA challenge, building a full pipeline under a tight deadline that fused over 5 data modalities (pathology, volumetric imaging, tabular) into a single cancer recurrence prediction system.
- Invited lecturer for the Data Engineering and Signal Processing courses, teaching 100+ students from varied backgrounds.
- Organized summer schools ([INVICTA 2025](#)), hosted 20+ internal invited talks with research and industry leaders in AI, and led "Paper of the Week" sessions to disseminate knowledge among peers.

Self-employed

Consultant

07/2024 – Present

- [Pocy.dev](#) (Apr 2026 to Present, pre-seed startup): designing a data evaluation pipeline for teams training humanoid robots; combining video and sensor data with VLMs to mine per-episode descriptions, and replacing qualitative spot checks with verifiable ground-truth events and scoring models that rank execution quality, not just task success.
- [Minute Audits](#) (Sep 2025 to Apr 2026, pre-seed startup): advised on an LLM agent platform (LangChain, LangGraph) automating the audit lifecycle, with agents handling pre-audit document requests and OCR/RAG pipelines grounding automated report filling against industry-specific norms.
- [iLoF](#) (Dec 2024 to Mar 2025, photonics biotech startup): scoped a vague "our analysis is unreliable" complaint into a concrete diagnosis, rebuilt the optical signal analysis pipeline, and added spatial and temporal stability tests validated against physical models before handing results to a non-technical clinical team.
- [Smartex.ai](#) (Jul to Oct 2024, computer vision startup): diagnosed mechanical and algorithmic flaws invalidating a camera supplier's qualification results, then redesigned the test jig and acceptance metrics, cutting camera rejection from 50% to under 5%; reported findings directly to the founding team.

Smartex.ai

Computer Vision Engineer

06/2023 – 06/2024

- Owned the full PoC lifecycle for a line-scan camera fabric inspection system (Master's thesis): scoped hardware and system requirements, built the acquisition and data pipeline, trained and evaluated the defect model, containerized the stack with Docker, and reported results to stakeholders. Delivered 8x inspection throughput (5 to 40 m/min), unblocking a product line previously limited to a single machine type.

Education Master of Science in Engineering Physics, 2024

University of Porto · 19/20 (Honors)

- Best Master's Student Award
- Thesis: Computer Vision System for Textile Inspection

Bachelor of Science in Engineering Physics, 2022

University of Porto · 18/20 (Honors)

Projects Marketplace Listing Auto-Tagging Platform

Human-in-the-loop system that replaces a manual tagging step for online marketplace listings: a scheduled scraper collects listings into PostgreSQL, a local vision-language model classifies category, condition, and attributes with structured outputs, and Label Studio provides the review interface. Reviewed listings become a labelled evaluation set that measures per-attribute accuracy and drives prompt selection and fine-tuning, closing the loop from human correction to model update. Fully containerized with Docker.

Scientific Research Agentic Platform

Agentic platform that answers research questions over scientific literature: a conversational interface routes a question to tool-using agents that plan multi-step searches across independent services for paper discovery (Semantic Scholar API), PDF download and OCR, embedding, and reranked retrieval, backed by Pinecone and PostgreSQL. Built to test how far a tool-using agent architecture scales on real retrieval workloads.

Automated Model Training and Deployment Pipeline

Infrastructure-as-code pipeline that takes a model from local training to a live cloud endpoint with no manual steps: MLFlow tracks training runs and a promote step pushes the selected weights to S3, Terraform provisions the AWS infrastructure, and a version tag triggers CI/CD (GitHub Actions builds the container, Ansible deploys it to EC2), where the service loads its weights from S3 at boot and serves them over FastAPI.

Languages Portuguese

Native speaker

English

Fluent

References

António Rocha - CTO, Smartex.ai
Jaime Cardoso - Researcher and Board Member, Faculty of Engineering, University of Porto